

Chapter 0: Introduction

Basics and Applications of Deep Learning

Fall 2026

Department of Mathematical Data Science

Hanyang University ERICA

Lecturer: Sehyun Kwon



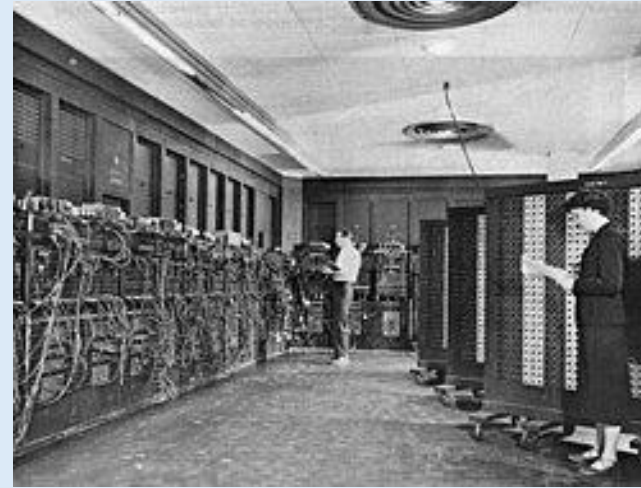
Humans build machines



A hand axe over 1 million years old.



Watt steam engine, 1776



ENIAC, 1946



Iphone 3, 2009

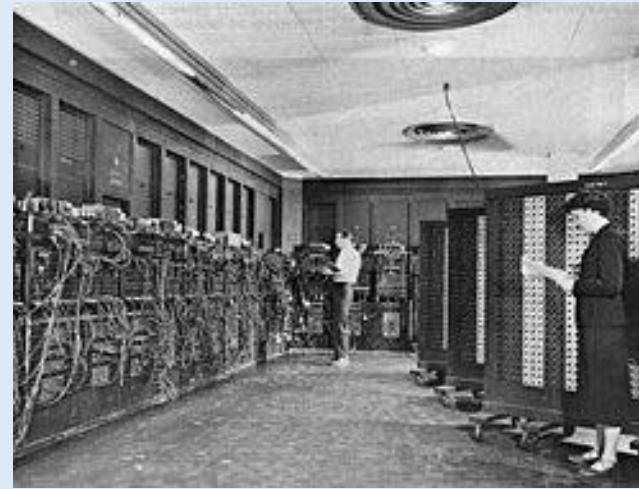
Humans build machines



A hand axe over 1 million years old.



Watt steam engine, 1776



ENIAC, 1946



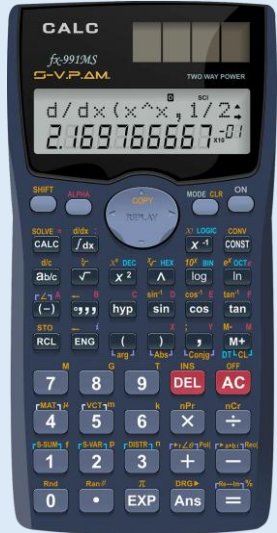
Iphone 3, 2009

But in the end, they are all machines that operate within **goals and rules defined by humans.** There is no intelligence.

What is *intelligence*?

There is no single clear definition.

But we recognize intelligence by its features: **learning**, **adaptation**, **reasoning**, and **generalization**.



Vs.

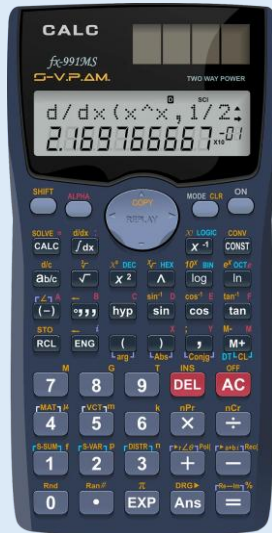


Which one is intelligent?

What is *intelligence*?

There is no single clear definition.

But we recognize intelligence by its features: **learning**, **adaptation**, **reasoning**, and **generalization**.



Vs.



A calculator is fast.
A baby is slow.

But only one of them learns.

Why do machines need *intelligence*?



Example: lane detection

Rule based solution (e.g., Hough transform):

1. Find bright lines.
2. Find edges.
3. Detect straight lines.
4. Call them lanes.

Why do machines need *intelligence*?



Rule based method is not scalable.



What if there are building edges?

What if poles or curbs look like lanes?

What if other markings confuse the system?

What if the scene is full of straight lines?

Solution: Learning from data



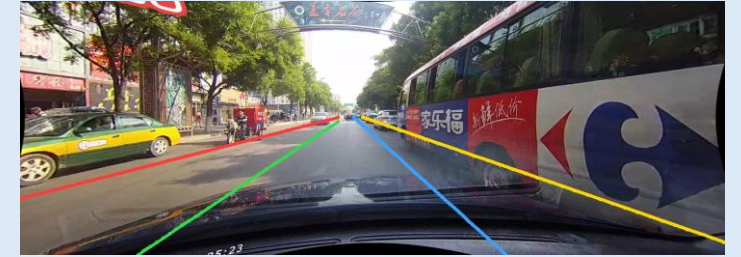
⋮



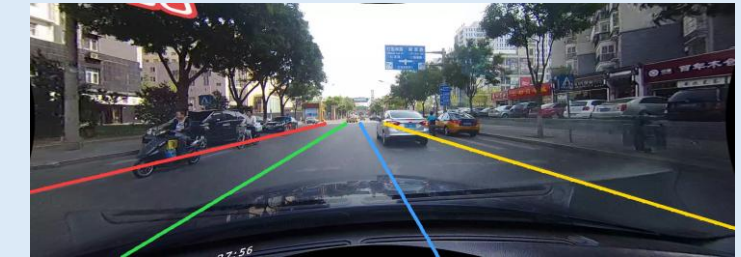
Many Many data

**Learning
Algorithm**

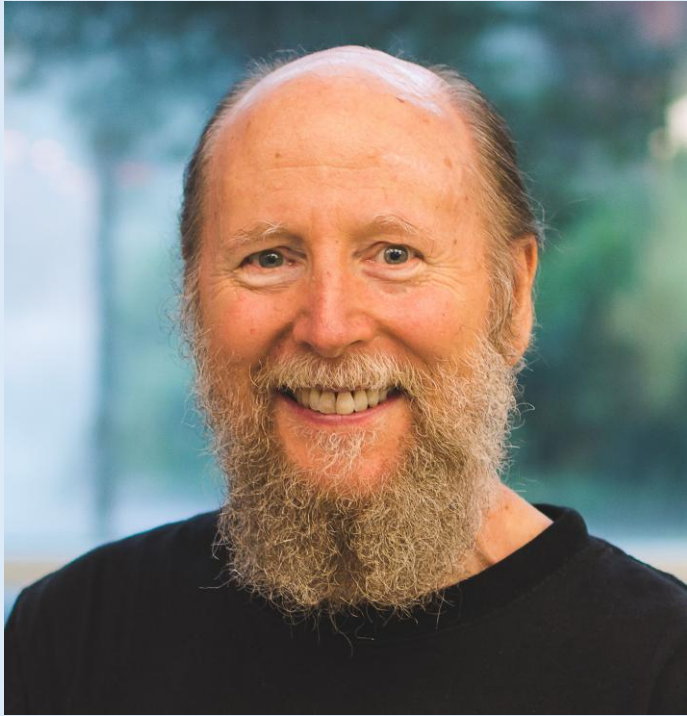
Instead of writing every rule,
we learn patterns from data.



⋮



The bitter lesson



Richard S. Sutton.

He is one of the pioneers of reinforcement learning and received the Turing Award.

“Simple architectures—backed by a generous computational budget, data set size and parameter count—surpass more complicated algorithms”

Bigger and bigger model

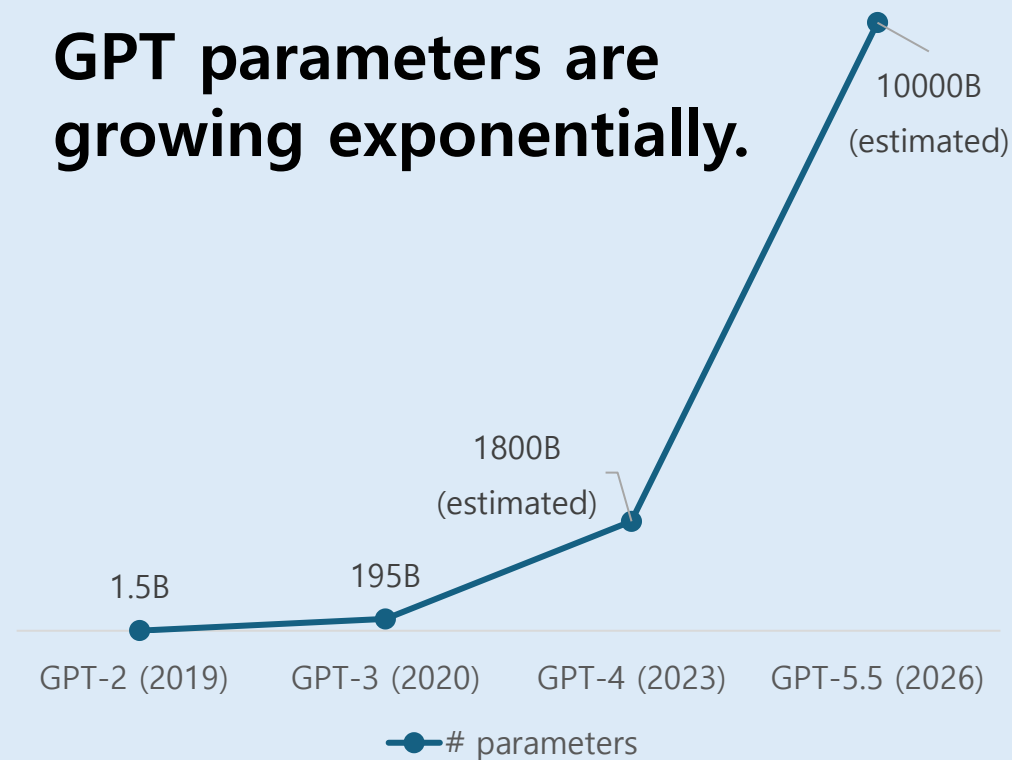
	A	B	C
1	1.1345	-0.123	3.182
2	1.414	1.171	-0.991
3	-2.666	3.814	0.101

Assume each cell size of 1cm x 1cm.



1 Football field = 60M (0.06B) parameters.

GPT parameters are growing exponentially.



→ GPT-5.5 size $\approx$ Seoul (650 km^2) + 2 * Ansan (156 km^2)

More and more data

2018 GPT 1 –600 Million Tokens (60 shelves)

2019 GPT 2 –28 Billion Tokens (2800 shelves)

2020 GPT 3 –300 Billion Tokens (30,000 shelves)

2023 GPT4 –1.3 Trillion Tokens (130,000 shelves = 650 km of shelves side by side)

1 library bookshelf = 10 million tokens



Deep Learning = GPU + Data + Model

Training:



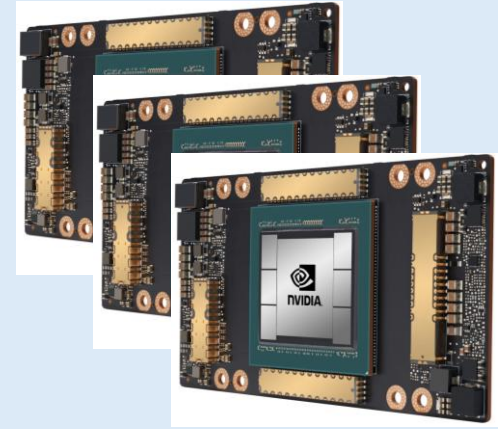
Data

+

model.safetensors Safe 459 MB xet

Model

+



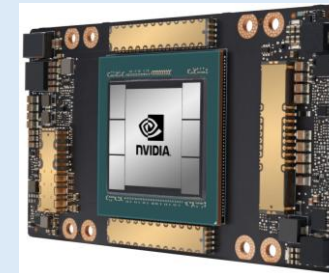
GPU

Inference:

model.safetensors Safe 459 MB xet

Model

+

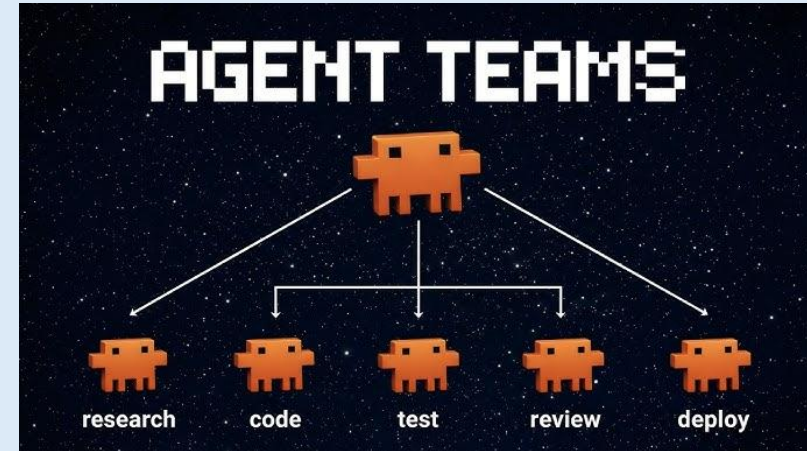


GPU

Applications of deep learning

There are so many applications. For examples:

Autonomous driving



Agent

chatbot

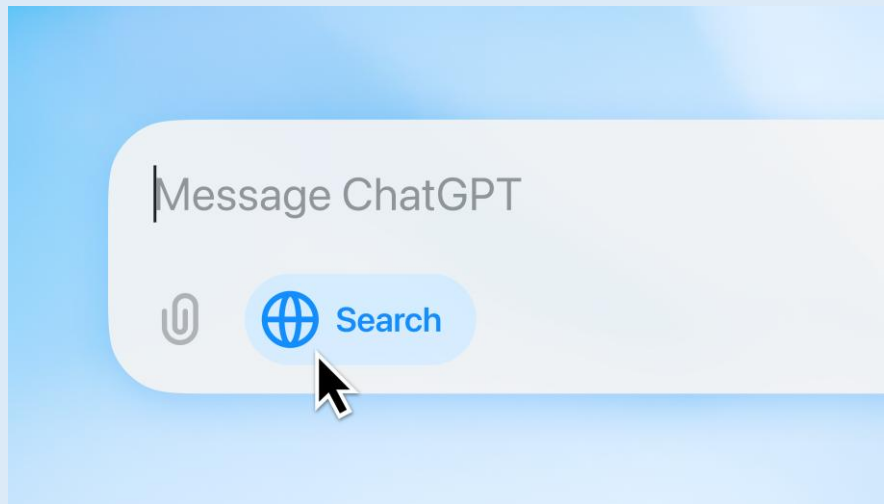


Image generation

